

PANDA Component Description: *bidir_port*

Overview

The *bidir_port* component implements a simple bidirectional parallel port with adjustable width. The port can be configured to synchronize the input data, which is used for asynchronous input signals, or not.

Generics

Name	Type	Default	Description
port_width	integer	8	Port width (min 1, max 8)
synchronize	boolean	true	true → Input is synchronized false → Input is not synchronized

External Ports

Name	Type	Width	Description
pio	std_logic_vector	<i>port_width</i>	Bidirectional Port

Registers

Output Register

Name: POUT

Address: Base + 0x00

Name	POUT
R/W	R/W
Reset Value	0x00

Each bit of the port holds the value of the corresponding bit in *POUT* if the corresponding bit in *TRIS* is not set. *POUT* holds the written value, even if this value does not reflect the real value at the port. To read the real port value, *PIN* must be used.

POUT has the same width as *pio* (*port_width*) and is right adjusted. If it is smaller than 8 bits, unused higher order bits are always read as '0' and can be written with any value.

Input Register

Name: PIN

Address: Base + 0x01

Name	PIN
R/W	R
Reset Value	-

The value at the port can be read through *PIN*. *PIN* has the same width as *pio* (*port_width*) and is right adjusted. If it is smaller than 8 bits, unused higher order bits are always read as '0'.

Tristate Register

Name: TRIS

Address: Base + 0x02

Name	TRIS
R/W	R/W
Reset Value	0xFF

If a bit in *TRIS* is set, the corresponding bit of the port becomes tristated and can be used as input. Otherwise it will be assigned the value of the corresponding *POUT* bit. After a reset, all bits are tristated.

TRIS has the same width as *pio* (*port_width*) and is right adjusted. If it is smaller than 8 bits, unused higher order bits are always read as '0' and can be written with any value.

Unused Register

Address: Base + 0x03

This Register is not used. Write operations to this register do not have any effect. Read operations result in a random result.